

Practical 02

Aim: To study and execute DML and DQL Commands.

Theory:

DATA MANIPULATION LANGUAGE (DML):

The Data Manipulation Language (DML) is used to retrieve, insert and modify database information.

These commands will be used by all database users during the routine operation of the database.

Let's take a brief look at the basic DML commands:

1. INSERT
2. UPDATE
3. DELETE

1. INSERT:

Values can be inserted into table using insert commands. There are two types of insert commands.

They are multiple value insert commands (using '&' symbol) single value insert command (without using '&' symbol).

Syntax:

```
INSERT INTO table_name VALUES (value1, value2, value3,.....);
```

(OR)

```
INSERT INTO table_name (column1, column2, column3,....) VALUES  
(value1,value2,value3,.....);
```

2. UPDATE:

This allows the user to update the particular column value using the where clause condition.

Syntax:

```
UPDATE <table_name> SET <col1=value> WHERE <column=value>;
```

3. DELETE:

This allows you to delete the particular column values using where clause condition.

Syntax:

```
DELETE FROM <table_name> WHERE <condition>;
```

DATA QUERY LANGUAGE (DQL)

DQL is used for querying and retrieving data from a database. It allows us to specify the exact data we want to see from one or more tables based on given conditions.

The SELECT command is used in Data Query Language Command.

SELECT:

The select statement is used to query a database. This statement is used to retrieve the information from the database. The SELECT statement can be used in many ways. They are:

1. Selecting some columns:

To select specified number of columns from the table the

Following command is used.

Syntax:

```
SELECT column_name FROM table_name;
```

2. Query All Columns:

To select all columns from the table * is used instead of column names.

Syntax:

```
SELECT * FROM table_name;
```

3. Select using DISTINCT:

The DISTINCT keyword is used to return only different values (i.e.) this command does not select the

duplicate values from the table.

Syntax:

```
SELECT DISTINCT column name(s) FROM table_name;
```

4. Select using IN:

If you want to get the rows which contain certain values, the best way to do it is to use the IN conditional expression.

Syntax:

```
SELECT column name(s) FROM table_name WHERE
```

```
Column name IN (value1, value2,.....,value-n);
```

5. Select using BETWEEN:

BETWEEN can be used to get those items that fall within a range.

Syntax:

```
SELECT column name FROM table_name WHERE
```

```
Column name BETWEEN value1 AND value2;
```

6. Select using JOIN:

Used to combine rows from two or more tables based on a related column between them.

- Types include INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN.

Syntax:

SELECT columns FROM table1 [JOIN TYPE] JOIN table2 ON table1.column_name = table2.column_name;

7. Select using GROUP BY:

Used with aggregate functions (like COUNT, MAX, MIN, SUM, AVG) to group the result set by one or more columns.

Syntax:

SELECT column1, aggregate_function(column2) FROM table_name GROUP BY column1;

8. Select using ORDER BY:

Used to sort the result set in ascending or descending order.

Syntax:

SELECT column1, column2 FROM table_name ORDER BY column1 [ASC|DESC], column2 [ASC|DESC];

Write down queries to be executed in practical slot with screen shot of output.

CONCLUSION:

The DML and DQL commands are executed successfully.